

VIBHAS TALLAPALLI

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EDUCATION

University of Waterloo

Candidate for BASc in Nanotechnology Engineering
Expected Graduation: May 2030

Waterloo, ON
Sep 2025 – Present

TECHNICAL SKILLS

Microcontrollers: Arduino, ARM, AVR, embedded sensors, serial communication, hardware wiring debugging, and circuit prototyping.

Mechanical & Manufacturing: SolidWorks, Onshape, CAD Modeling, Tolerances, GD&T, DFM, DFA, Carbon-Fiber Fabrication, 3D Printing, Machining (Mill, Bandsaw, Lathe), Assembly, tolerance analysis.

Software: C++, Python, C, Arduino (C++), MATLAB, Git, Java, PID

EXPERIENCE

Waterloo Design Team (University of Waterloo)

Mechanical Engineering Trainee

Waterloo, ON
Sep 2025 – Present

- Owned and maintained a **100+ part Onshape assembly** integrating brakes, suspension, and battery packaging, used by the mechanical team as the **primary reference**.
- Designed brake mounting plates and pads using **GD&T**, achieving **first-pass manufacturing** and installation by a **20** member manufacturing team.
- Defined braking system interfaces (dimensions, clearances, hole locations) used across a **30+** member mechanical team.

Electrium Design Team (University of Waterloo)

Mechanical Engineering Trainee

Waterloo, ON
Sep 2025 – Present

- Created the **first SolidWorks CAD** of the e-bike frame, enabling battery enclosure and firmware packaging for a **35** member mechanical team.
- Led battery enclosure material selection, replacing aluminum with tempered glass and **saving \$300+** across multiple builds.

Biomechatronics Design Team (University of Waterloo)

Mechanical Engineering Trainee

Waterloo, ON
Sep 2025 – Present

- Prototyped and tested thigh and shin strap systems for a first-generation lower-body exoskeleton using **CAD** layouts and **3D-printed** fixtures, stabilizing **4** human–device attachment points during early system development.
- Built and tested **2** strap-routing prototypes in **Fusion 360**, preventing frame interference and leg motion restriction.

PROJECTS

Smart Study Focus Timer

Python, SolidWorks, Arduino

Winter 2026

- Designed and built an embedded sensor device using **3 optical distance sensors** and an Arduino Uno, achieving **95% reliable presence detection** after **25+** hardware and logic iterations.
- Expanded detection range **45% beyond a laptop webcam**, enabling single-user and group use (**20+ users**) through sensor tuning, timing logic, and a custom enclosure.
- Implemented embedded control logic in **Arduino (C++)**, managing sensor polling, timing thresholds, and LED states to support **95% reliable** operation.

E-Bike Speed Control Simulation

C++, Python

Fall 2025

- Tuned **PID** speed control for an e-bike under load and terrain changes, improving simulated energy efficiency by **33%** and stability.
- Optimized a PID controller to reduce speed overshoot by **15%**, settling time by **20%**, and improve simulated energy efficiency by **40%** under changing load conditions.

Energy Collecting Turbine

CAD, Mechanical, Electrical

Fall 2025

- Built and CAD-modeled a rotating turbine component, increasing measured **DC output voltage by 32%** through iterative design and testing.
- Converted turbine rotation into **DC electrical output** by driving a generator during wind testing and measuring output voltage with a **digital multimeter**.